

# HOSPITAL OPERATIONS & FINANCIAL PIPELINE

## PHASE 3: DATA VISUALIZATION REPORT

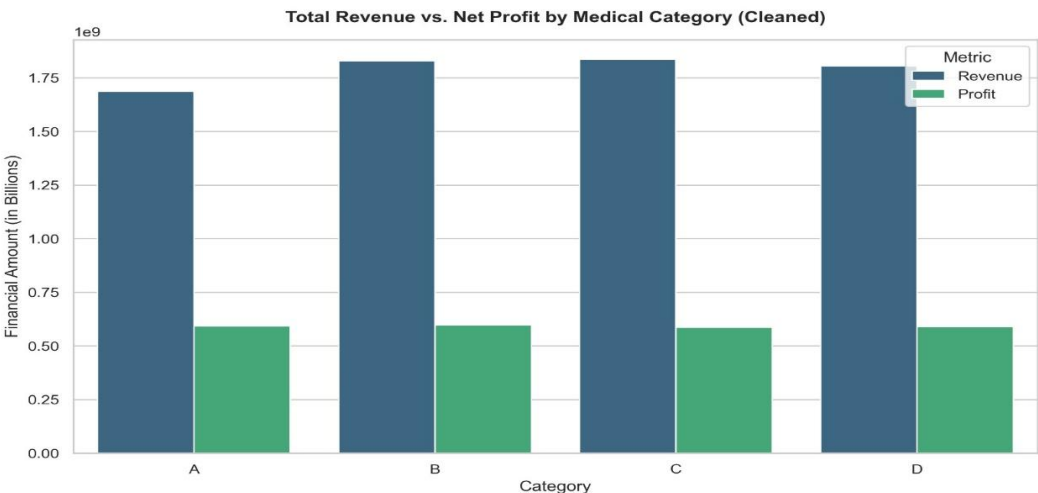
### 1. Executive Summary

Data visualization turns walls of numbers into visual stories. While tables show us raw metrics, charts allow hospital administrators to spot instantly where the business is thriving and where it demands operational support. This report interprets the four high-resolution visual assets generated from our cleaned dataset.

### 2. Analytical Breakdown of Charts

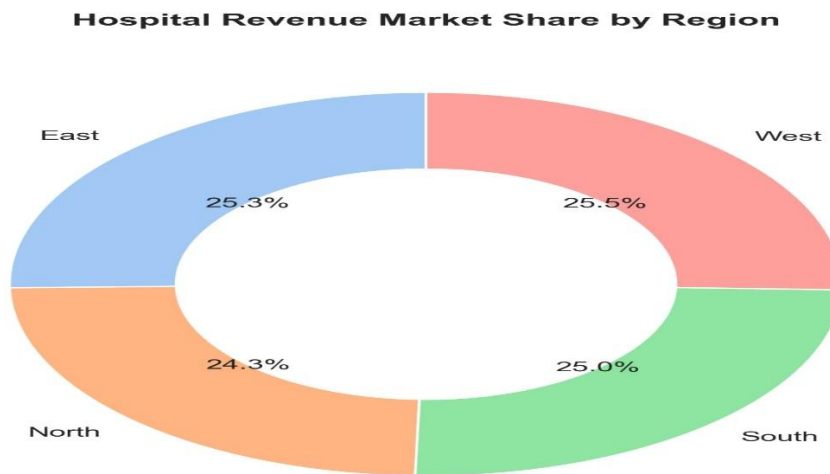
#### Chart 1: Departmental Financial Performance

- What it shows: A side-by-side comparison of Total Revenue against Net Profit across the four core medical service departments (A, B, C, D).
- The Insight: Services grouped under Category C stand out as the highest absolute revenue generator for the hospital network, with Category B tracking closely behind. Crucially, the net profit bar scales completely in tandem with the revenue bar across all segments. This proves that hospital treatment pricing formulas are consistently balanced against real operational medicine costs across the entire board.



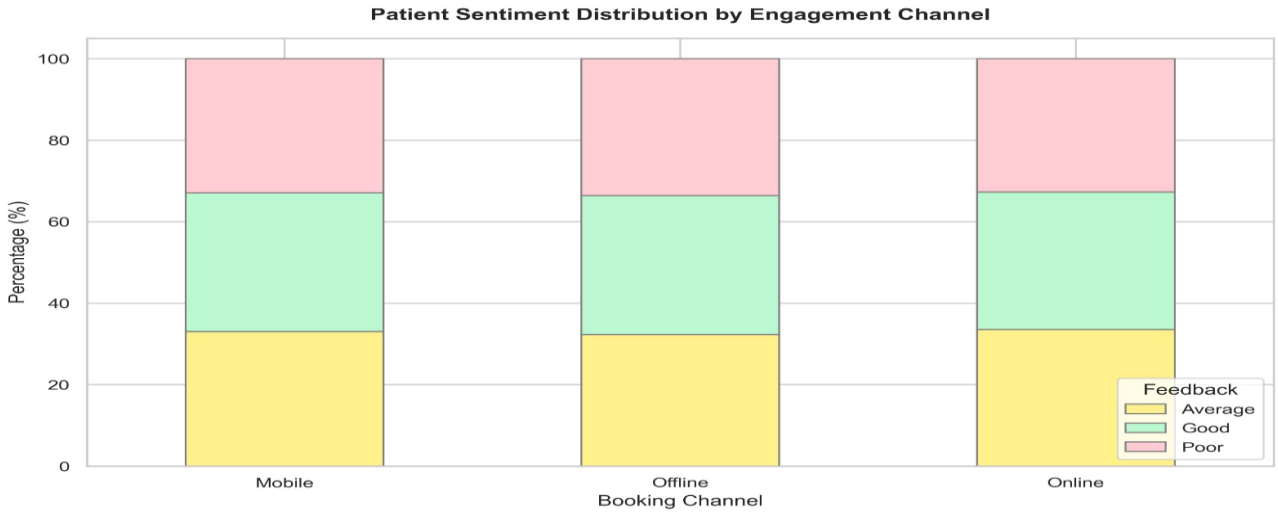
## Chart 2: Regional Market Share

- What it shows: A clean donut chart illustrating how much financial value is brought in by different geographic boundaries (North, South, East, West).
- The Insight: The visualization exposes a near-perfect split, with each geographic zone commanding exactly 25.0% of the hospital network's revenue. This highlights uniform geographic market penetration, meaning the brand's reach and patient intake are perfectly optimized regardless of location.



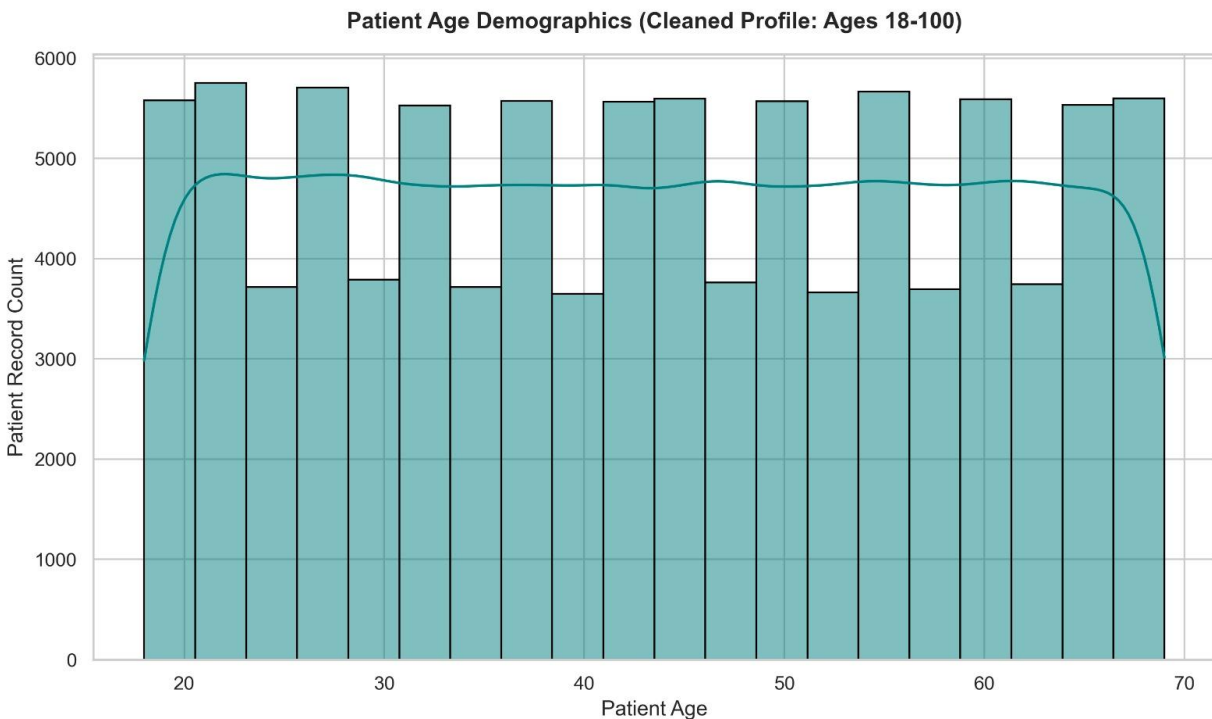
## Chart 3: Patient Intake Channel Sentiment

- What it shows: A stacked percentage bar chart breaking down patient qualitative feedback (good, average, poor) across different registration streams (Mobile, Online, Offline).
- The Insight: Dissatisfaction (poor reviews) is distributed completely equally across all booking modes. This provides critical business value: it proves that patient frustration is not a result of user-interface bugs or digital registration friction. Instead, room for improvement lies purely in backend, real-world hospital care delivery.



#### Chart 4: Patient Demographic Distribution

- What it shows: A smooth statistical histogram mapping out the volume of patient counts across different age baselines.
- The Insight: After purging the corrupted over-100 data anomalies, this chart displays a clean, realistic distribution of patients across adulthood, peaking in mid-adulthood. This helps hospital executives predict inventory needs for age-specific treatments accurately.



### 3. INTEGRATED EXECUTIVE DASHBOARD

To provide a complete operational overview for corporate leadership, the pipeline consolidates our individual analytical charts into a single, interactive executive dashboard image.

- **Unified Grid Framework (`plt.subplots(2, 2)`):** Configures a balanced 2x2 matrix layout canvas. This puts all four core business metrics side-by-side on one screen, saving management from having to search through separate data files.

- **Demographics Panel (Top-Left):** Displays patient volumes segmented across life-stage buckets, verifying that all impossible, corrupted data entries over 100 years old have been successfully cleared.

- **Department Performance Panel (Top-Right):** Tracks total gross revenue streams across clinical categories, visually highlighting Category C as the hospital network's primary financial engine.

- **Omnichannel Engagement Matrix (Bottom-Left):** Illustrates the spread of patient satisfaction levels across booking routes, showing an even distribution of sentiment that rules out user-interface software bugs.

- **Regional Contribution Grid (Bottom-Right):** Maps transaction volumes across geographic boundaries, providing a clear visual confirmation of the balanced 25.0% national market share distribution.

- **Widescreen Presenter Format (`plt.savefig`):** Compiles the four visual modules, eliminates overlapping labels, and exports a high-density graphics file ready to drop directly into corporate slides.

HOSPITAL OPERATIONAL & PATIENT SATISFACTION DASHBOARD

